

Prof. Christopher John Budd OBE, FAcadMathSci, FIMA, C.Math, NTF, FGC

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Present and recent appointments

1995-Present. Professor of Applied Mathematics, University of Bath,
 2020-Present. Director of Knowledge Exchange at the (Bath) Institute for Mathematical Innovation (IMI),
 2026-Present. Fellow of the Academy for the Mathematical Sciences
 2022-Present. Principal Investigator for the UKRI Programme Grant: Maths4DL,
 2023-Present. KE-Hub Super Champion, and executive member
 2000.-Present. Chair of Mathematics at the Royal Institution of Great Britain.
 2022-Present. Fellow of Gresham College,

Previous appointments

2016-2019. Gresham Professor of Geometry,
 2000-2022. Chair of the (Bath) Centre for Nonlinear Mechanics.
 2004-2010. Director of the Bath Institute for Complex Systems.
 1989-1995. Lecturer/Reader in numerical analysis, University of Bristol.
 1986-1989. CEGB Research Fellow in numerical analysis, Hertford College Oxford.

Academic qualifications

1983-1986. D. Phil. in Mathematics, Oxford University, 1983-1986.
 1982-1983. Part 3, Cambridge, 1982-1983, Distinction.
 1979-1982. Mathematics M.A. Degree, Cambridge University, First Class Honours.

Research Interests: I have been an active research mathematician since 1983. My original work in nonlinear PDEs has since developed in many directions. I have done much work in non-smooth dynamics, and co-authored what has become the standard text book for the field. I have developed effective methods for adaptive mesh generation which are now used operationally by the Met Office and greatly reduce errors in near ground temperature forecasts. I have done much work in climate dynamics, including collaborating with African mathematicians on the impact of climate change on agriculture. My recent work is in Scientific Machine Learning, for which I run a large research group. I have organised many research conferences including BAMC, SciCADE, and Study Groups. The recent Scientific Machine Learning conferences I organised have been deliberately open to involve the whole community.

Teaching and Education: I have taught many courses in applied maths/numerical analysis. I set up the Bath MSc in Applied Mathematics in 2000, involving a strong set

of industrial collaborators. Following the award of a HEA National Teaching Fellowship in 2001 I set up the Communicating Maths course, teaching the methods of mathematics communication to third year undergraduates who then deliver outreach at RI maths masterclasses, schools, science festivals etc. Course alumni have gone on to be leading maths communicators. Similar courses have now been set up in other universities. I am a strong supporter of mathematics education. I served as LMS Education Secretary, and on the 2010-11 Vorderman Ctee advising HM Govt on maths education. A recommendation of this Ctee was that maths should be taught for all ages up to 18.

Knowledge Exchange and Public engagement: I was one of the founders of V-KEMS, worked closely with SPI-M-O during the COVID pandemic and received a governmental award for my work. I have served as KE Director of the Bath IMI for 10 years from 2015, and have seen it grow from small beginning to being a significant force in operational KE. I have been involved in the KE-Hub from the start and have been an active member ever since as a Super Champion. I set up the Bath MSc in 'Modern applications of mathematics' in 2001 and work with very many companies through this, setting up and coordinating over 300 industrially focused projects. I am a leading international proponent of mathematically based **Public Engagement** and my work in this field has been recognised by the award of an OBE. In particular I set up the Bath Science Festival, the Bath Communicating Maths course, have written three popular maths books, run the Bath Royal Institution maths masterclass programme, have written many articles, and have given many talks popularising maths. Have worked with many **policy makers** including visits to Westminster and Parliament.

Community service Fellow of the Academy of the Mathematical Sciences, IMA Council and Vice-President, IMA Research Committee, IMA Special interest group on climate, environment, and sustainability (co-chair), executive and founder member of V-KEMS, KE Hub super champion and working group on AI (co-chair), LMS Council, LMS Education Secretary, member of the Vorderman Ctee 2010-11, REF2014 Panel, Royal Society RAMP ctee 2020-2021, UKMT Chair 2016-2019.

Editorships IMA Transactions of Mathematics (lead editor), Royal Society Open Science (associate editor)

Conference organisation: Lead organiser, or on the executive committee of many conferences, including most recently: 2015 CliMathNet (100 participants), 2016 Moving Mesh Methods (100 participants), 2017 SciCADE (400 participants), 2018 ESGI (80 participants), 2019 BAMC (350 participants), 2020 Mathematical Models for Weather and Climate Prediction (160 participants), 2022 ICMS workshop on adaptive mesh methods, 2023 ESGI (80 participants), 2023 BMC, as well as many Maths4DL conferences, workshops and hackathons (at least three per year) from 2022-present. I have been lead organiser of 4 ESGI study groups and 15 Virtual study groups.

Postgraduate supervision: PhD Students: 42 completed, 5 current, 20+ Post Doctoral Research Assistants. Over 40 MSc students in industrial applied mathematics.

Grants: Principal Investigator on research grants totaling over £7M including a £1M

EPSRC Critical mass grant on Complexity for 8 PDRAs (2004), a Marie-Curie ITN Grant on PDES for 39 PhD students (2009), a NERC grant for 4 PDRAs on moving mesh methods (2015), £3.5M EPSRC Programme Grant (Maths for Deep Learning 2022-2027), 5 Innovate UK A4I Grants (2023-2025). Principal investigator on public understanding of science grants totaling over £200k including Evidence Week (2025), Bath Taps Into Science 2000-present, and the 2010 Royal Society Summer Exhibition.

Publications: I have written 4 books, and over 160 journal/conference publications. H-index 43. For details see <https://researchportal.bath.ac.uk/en/persons/chris-budd>

Awards, honours and distinctions

- 1982. Senior Wrangler (top first class degree), Cambridge.
- 1991. First Prize in the international Leslie Fox competition for Numerical Analysis.
- 1999. Elected one of ten ‘Scientists for the New Century’ by the Royal Institution.
- 2001. ILT/HEA National Teaching Fellowship (NTFS).
- 2001. LMS Popular Lecturer in Mathematics.
- 2002. IOP Award for Outstanding Contributions to the Public Understanding of Physics.
- 2009. British Science Association Prize for the Best Science Festival in NSEW.
- 2011. Honorary Fellow of the British Science Association.
- 2012. University of Bath/Met Office Knowledge Transfer Award.
- 2015. Seelye Distinguished Fellowship, Auckland.
- 2015. IMA Distinguished Service Award.
- 2015. OBE for work in Mathematics and Science Education.
- 2015. LMS Prize for Communicating Maths.
- 2020. Joint JPBM AMS/MAA/SIAM prize for communicating mathematics.
- 2021. University of Bath award for leadership in public engagement.
- 2021. Praxis-Auril Prize for Knowledge Exchange.
- 2021. University of Bath Chancellor’s Research Medal.
- 2022. University of Bath prize for PhD supervision.
- 2022. SPI-M-O SAMDS award for Modelling and Data Support.
- 2025. David Rees Prize Fellowship (University of Exeter).
- 2026. Fellow the Academy for the Mathematical Sciences
- 2026. INI (Cambridge) Simons Fellowship
- 2026. IMA Gold Medal